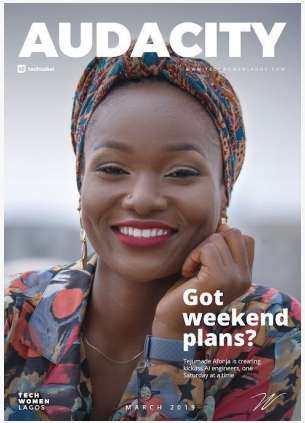
Mathematics for Machine Learning

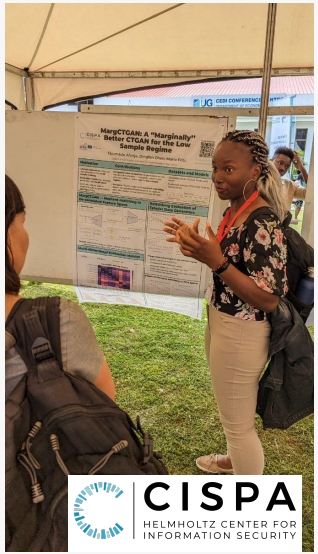
IndabaXUganda

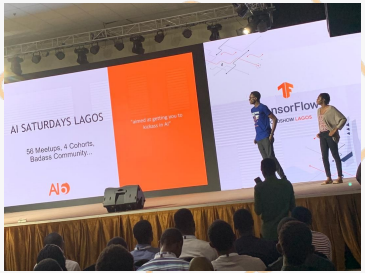
Instructor: Tejumade Afonja

23.03.2026

About Me

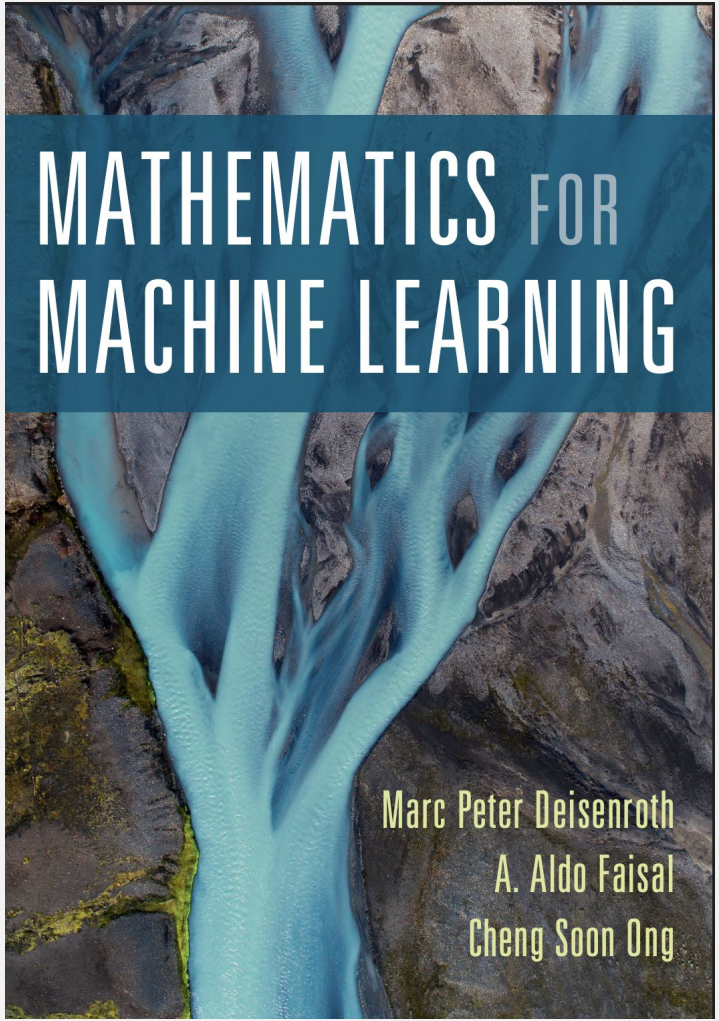






- Executive Director & Co-founder, TRI AI
  - Co-founder, AI Saturdays Lagos
- PhD Student, CISPA Helmholtz Center for Information Security
- Programme Committee Chair, Deep Learning Indaba 2026

What is Machine Learning?


MATHEMATICS FOR
MACHINE LEARNING
Marc Peter Deisenroth
A. Aldo Faisal
Cheng Soon Ong

AI
‹#›

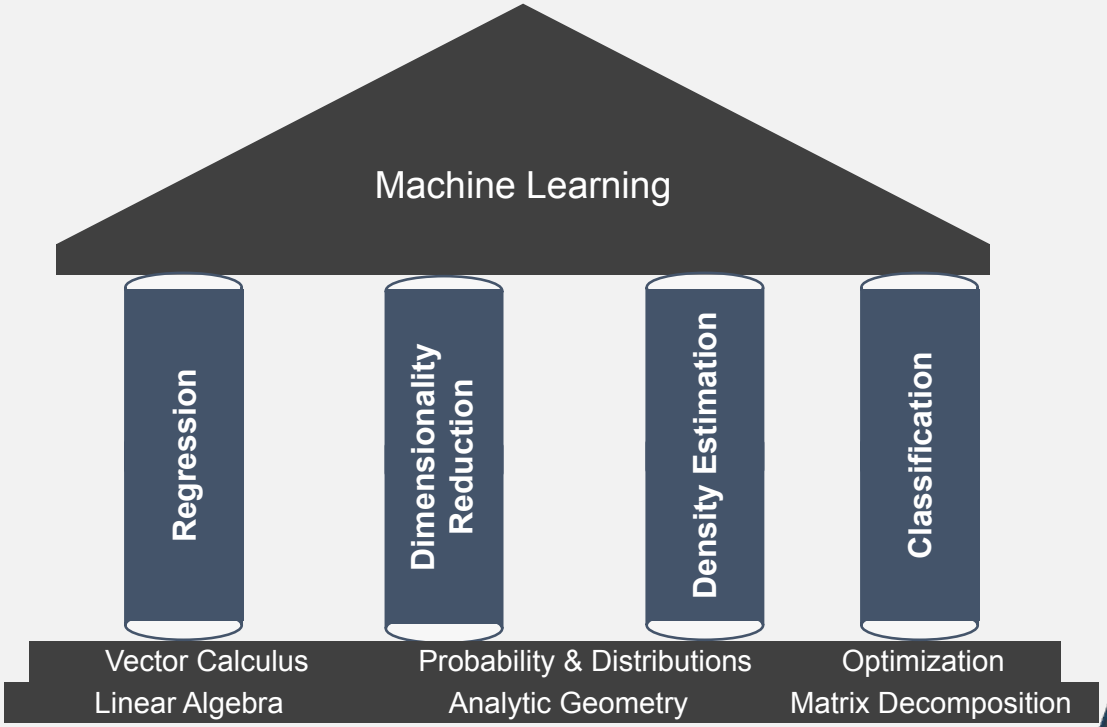


Figure adapted from the mathematics of machine learning textbook

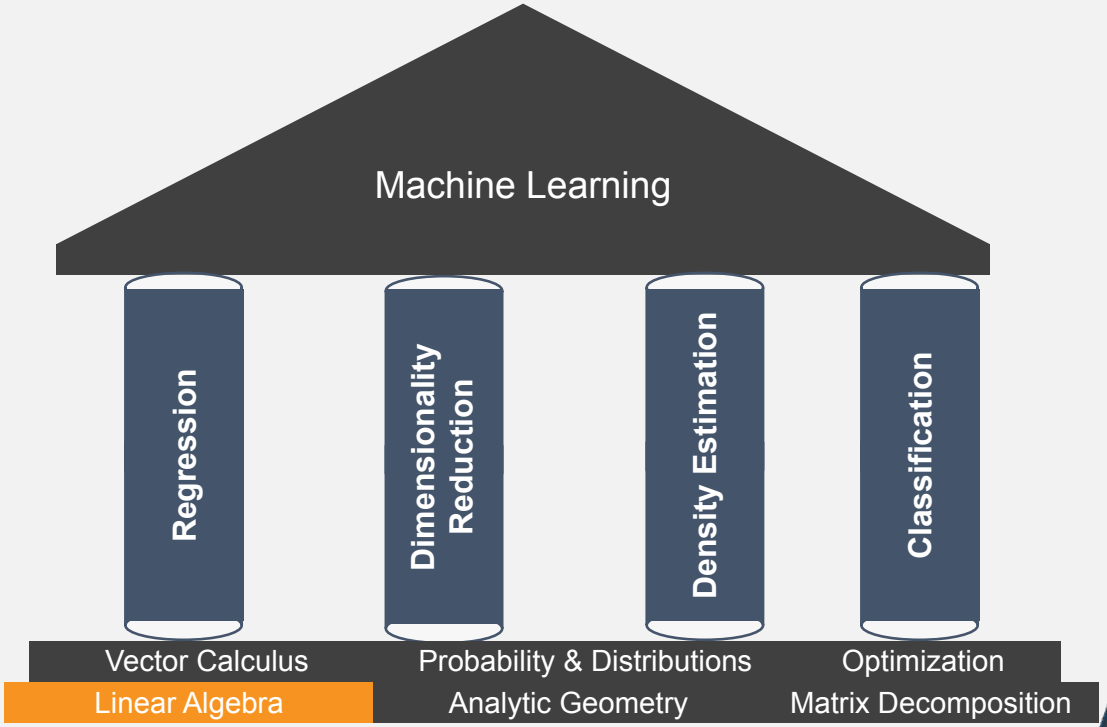


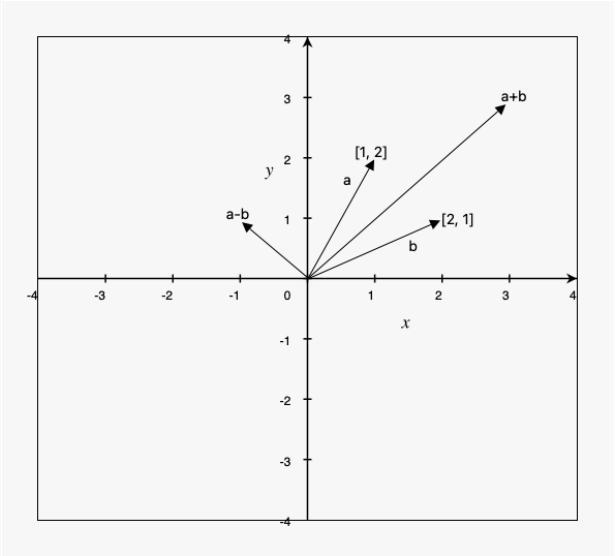
Figure adapted from the mathematics of machine learning textbook

Linear Algebra

The study of **vectors** and certain rules to **manipulate** them

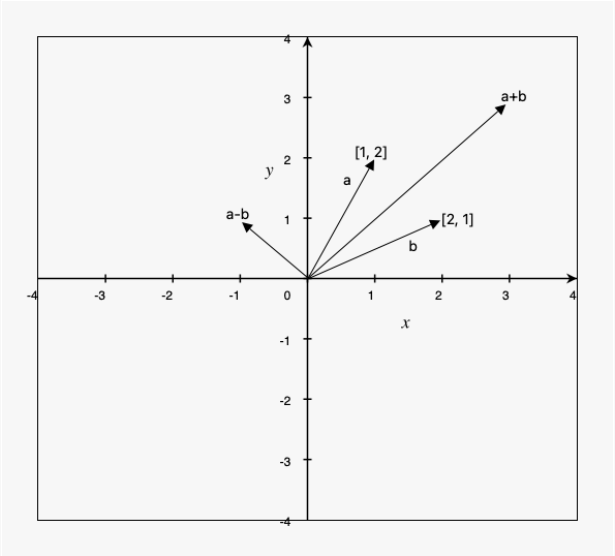
Linear Algebra

The study of **vectors** and certain rules to **manipulate** them



Linear Algebra

The study of **vectors** and certain rules to **manipulate** them

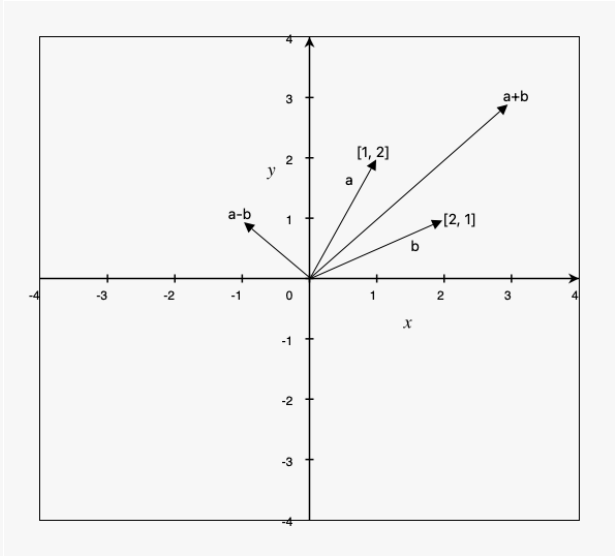


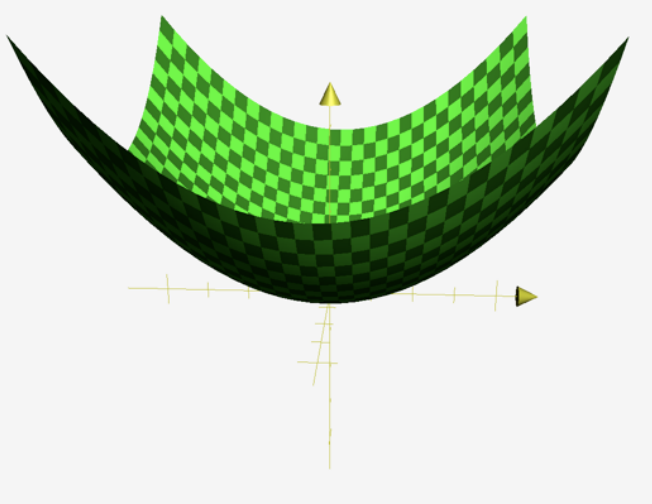
Vectors are special objects. Can be:

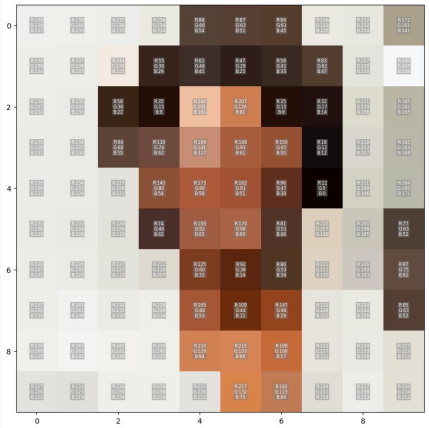
- Added
- Subtracted
- Multiplied by a scalar (a scalar is a number)

Linear Algebra

The study of vectors and certain rules to manipulate them







How many dimension does the column vector object has?

Each row and column represent a vector

Tabular Data

columns = attributes for those observations

Rows = observations

Player	Minutes	Points	Rebounds	Assists
A	41	20	6	5
B	30	29	7	6
C	22	7	7	2
D	26	3	3	9
E	20	19	8	0
F	9	6	14	14
G	14	22	8	3
I	22	36	0	9
J	34	8	1	3

Scalar, Vectors and Matrices

Scalar

$$24$$

Vector

$$\begin{bmatrix} 2 & -8 & 7 \end{bmatrix}$$

row

or column

$$\begin{bmatrix} 2 \\ -8 \\ 7 \end{bmatrix}$$

Matrix

$$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$$

row(s) × column(s)

Matrix are vectors stacked in rows or columns

A simplified (machine) learning problem

Imagine you want to predict house prices based on size.

	House size (m^2)	Price (100,000 USh)
1	10	200
2	20	400
3	50	1000

House size is the *feature* (x)

Price is the *target* (y)

Representing your data: scalar

Scalar is a single number: for example the price and size of the first house

House size (m^2)	Price (100,000 USh)
10	200
20	400
50	1000

$$x_1 = 10$$

$$y_1 = 200$$

Representing your data: vector

Vector is a list of numbers, usually use to represent the features and target

House size (m^2)	Price (100,000 USh)
10	200
20	400
50	1000

$$x = \begin{bmatrix} 10 \\ 20 \\ 50 \end{bmatrix} \quad y = \begin{bmatrix} 200 \\ 400 \\ 1000 \end{bmatrix}$$

Vector stack across many observations

Representing your data: matrix

Matrix is a rectangular grid or array of numbers, symbols, or expressions arranged in rows and columns.

House size (m^2)	Price (100,000 USh)
10	200
20	400
50	1000

We can represent the dataset as a matrix of feature (x) and target (y).

$$\text{Data}(\mathcal{D}) = \begin{bmatrix} 10 & 200 \\ 20 & 400 \\ 50 & 1000 \end{bmatrix}$$

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$$

This is a 3x2 matrix i.e 3 rows and 2 columns

Extending to multiple features

Now, suppose we add another feature, number of rooms.

House size (m^2)	Number of Rooms	Price (100,000 USh)
10	2	200
20	5	400
50	10	1000

Extending to multiple features

Now, suppose we add another feature, number of rooms.

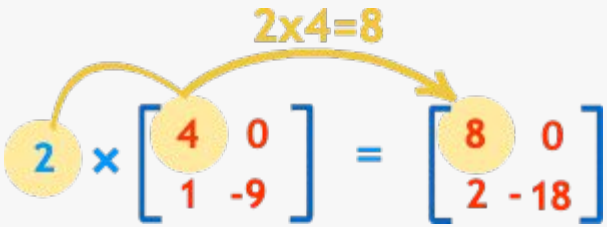
House size (m^2)	Number of Rooms	Price (100,000 USh)
10	2	200
20	5	400
50	10	1000

Now we present each feature as a matrix (X)

$$X = \begin{bmatrix} 10 & 2 \\ 20 & 5 \\ 50 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 200 \\ 400 \\ 1000 \end{bmatrix}$$

Some operations on matrices

Multiplying matrix by a scalar



https://www.mathsisfun.com/algebra/matrix-multiplying.html

Some operations on matrices

Matrix Addition

These are the calculations:

3+4=7	8+0=8
4+1=5	6-9=-3

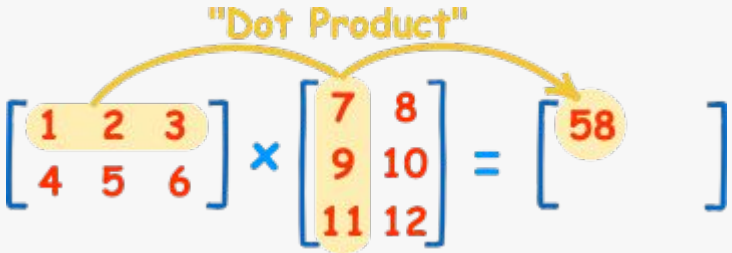
https://www.mathsisfun.com/algebra/matrix-multiplying.html

Some operations on matrices

Multiplying matrix with by another matrix

- Dot product

The number of columns in the first matrix has to match the number of rows in the second matrix



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$$c_{i,j} = \sum_{l=1}^{n} a_{il} b_{lj}$$

i, j is used to represent row, column

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Multiplying matrix with by another matrix

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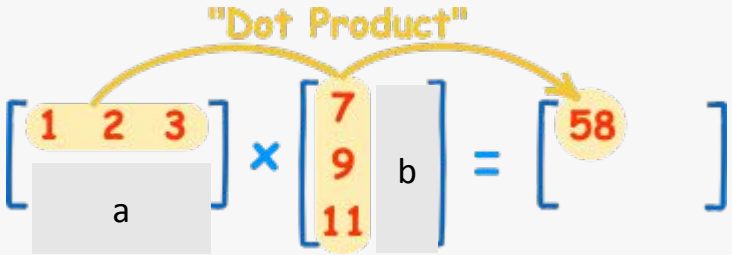

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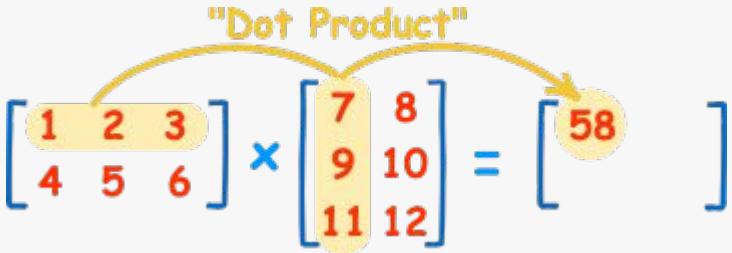
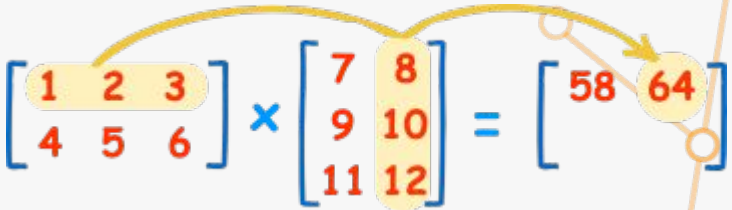




$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix} \checkmark$$

$$\underbrace{A}_{n\times k}\ \underbrace{B}_{k\times m} = \underbrace{C}_{n\times m}$$

https://www.mathsisfun.com/algebra/matrix-multiplying.html

Some operations on matrices

Transposing a matrix

Transpose of a Matrix

cuemath
THE MATH EXPERT

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2 \times 3} \qquad A^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3 \times 2}$$

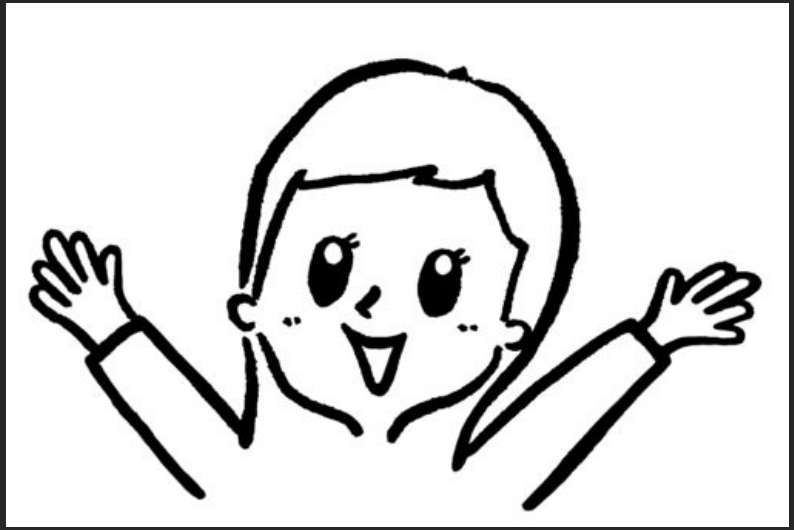
https://www.cuemath.com/algebra/transpose-of-a-matrix/

Vector and matrices practise Notebook

Practise Notebook:

https://colab.research.google.com/drive/1Jy3pSrHwzD-9sf8WwwhRG1L_CQ9-WsCR

Linear Algebra in Practice



Challenge

Imagine you want to predict house prices based on size.

House size (m^2)	Price (100,000 USh)
10	200
20	400
50	1000

Given a new house size of $80m^2$,
What is your predicted price

?

Your linear model

Your model

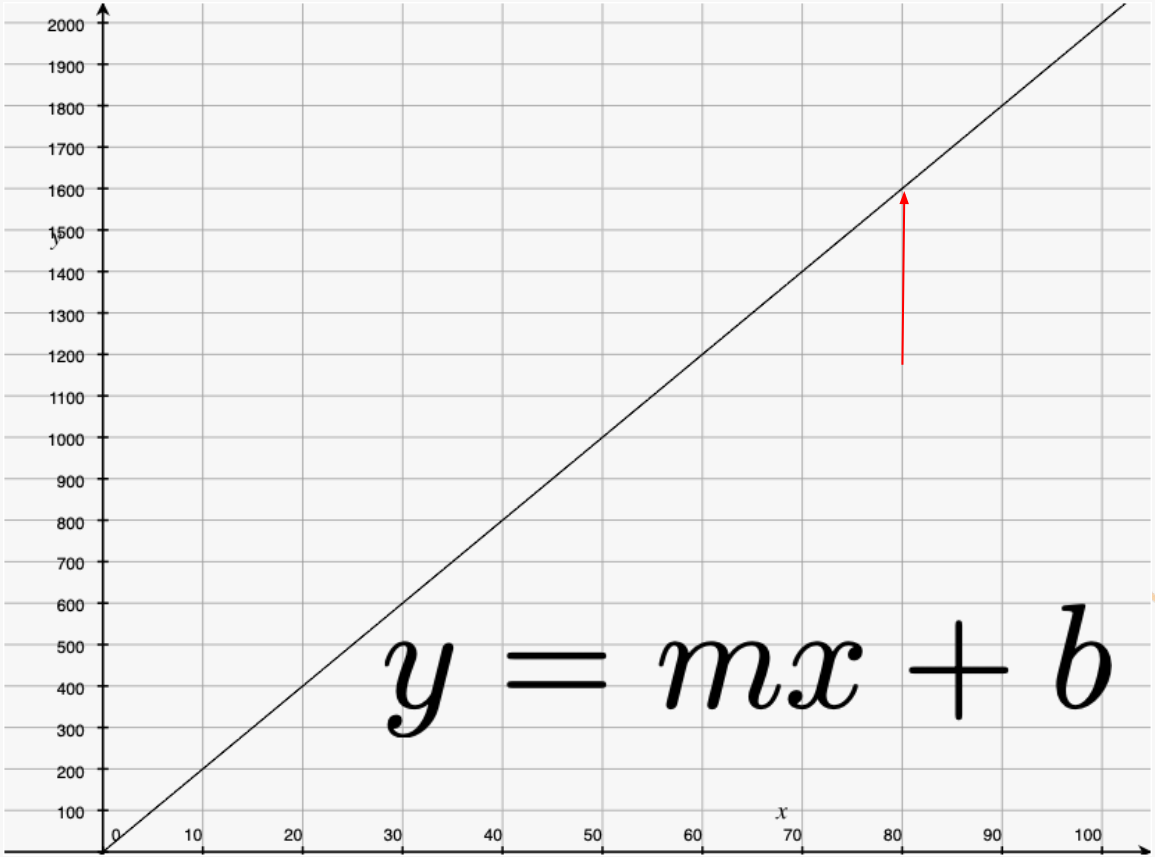
$$y = 20x$$

Assumes the relationship between your input x and output y is **linear**

Equation of a straight line

m is the slope / gradient(how much price *changes*)

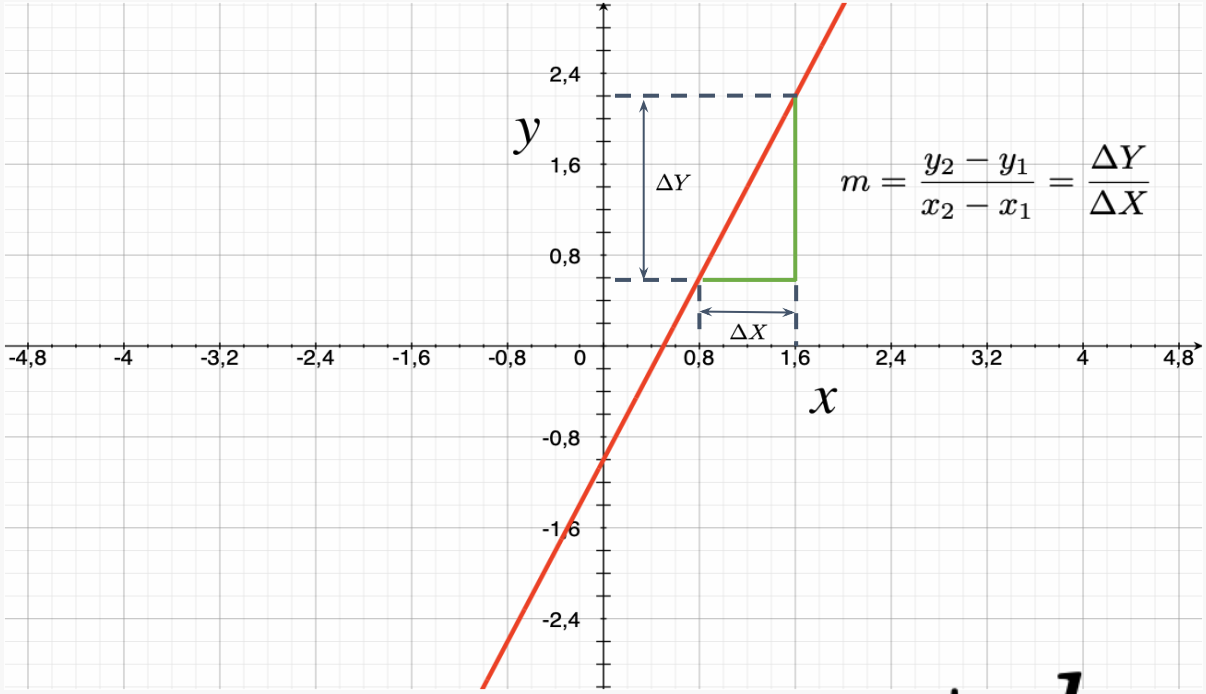
bias (b) is the *base price*



Your model equation

$$y = 20x$$

Calculating the slope / gradient of a linear function

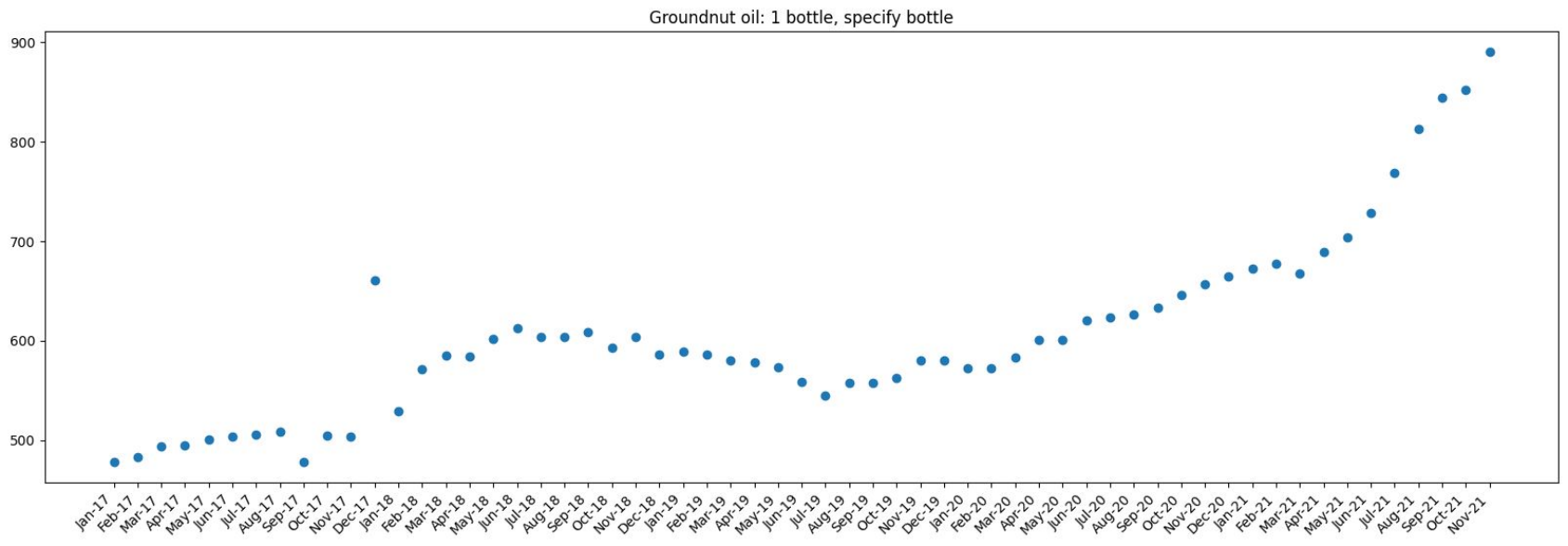


$$y = mx + b$$

?

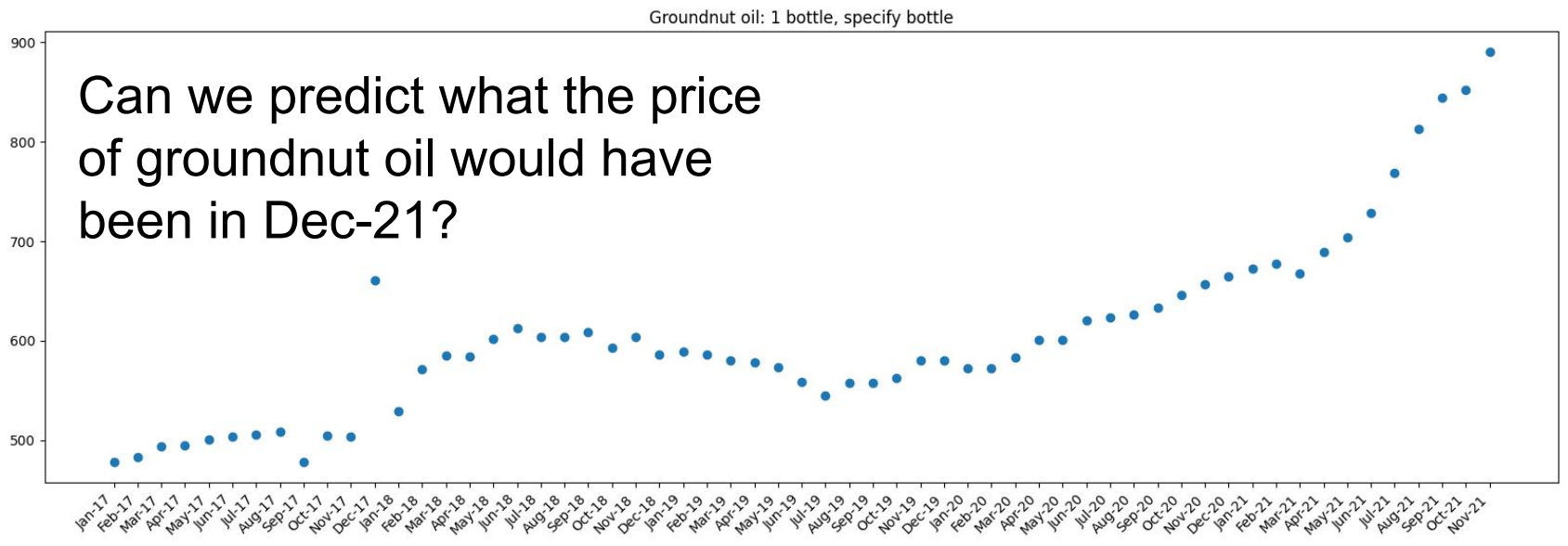
What is value of **m** and **b**

Real life data: Nigerian food item from 2017-2021



https://open.africa/dataset/food-prices-2017-2021

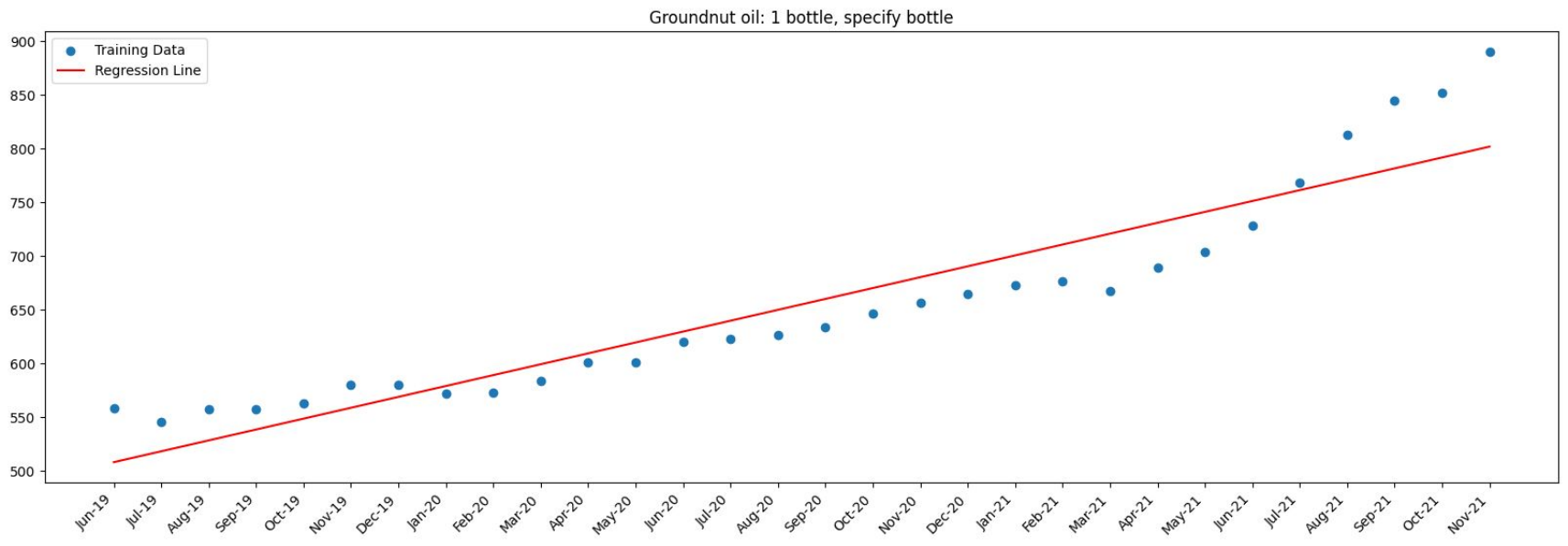
Real life data: Nigerian food item from 2017-2021



Can we predict what the price of groundnut oil would have been in Dec-21?

https://open.africa/dataset/food-prices-2017-2021

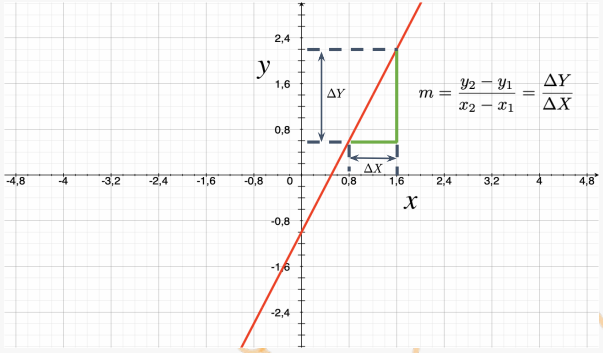
Fitting a straight line: linear regression



Linear Regression

Linear Regression assumes that there is a **linear** relationship between the features and the numerical label.

$$y = mx + c$$



In ML, we usually write the equation as:

$$y = b + w_1 x_1$$

where:

y is the label

b is the bias (y-intercept), sometimes written as w_0

w_1 is the weight of feature x_1

Linear Regression

The linear regression model predicts an output $\hat{y}$ based on input features $x_1, x_2, \cdots, x_n$.

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

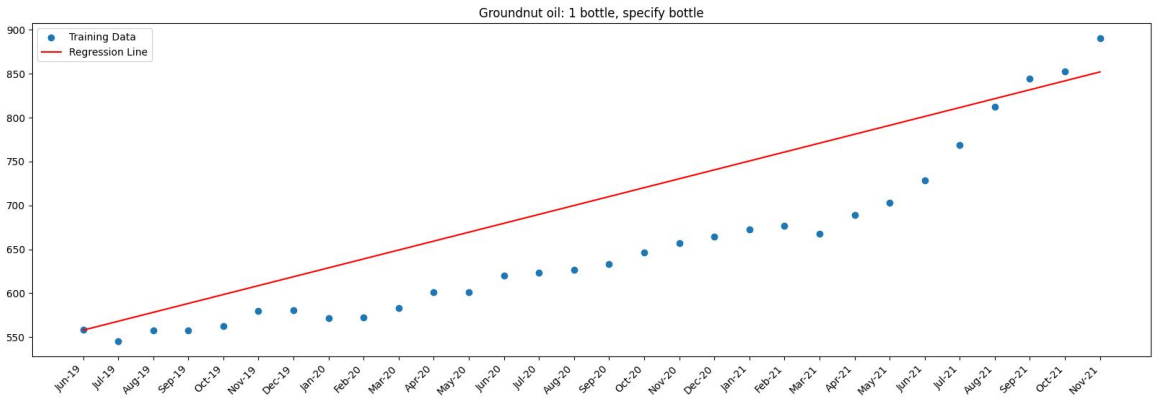
where:

- $\hat{y}$ is the prediction
- w_0 is the y-intercept (bias)
- $w_1, w_2, \cdots, w_n$ are the weights (coefficients)
- $x_1, x_2, \cdots, x_n$ are the input features

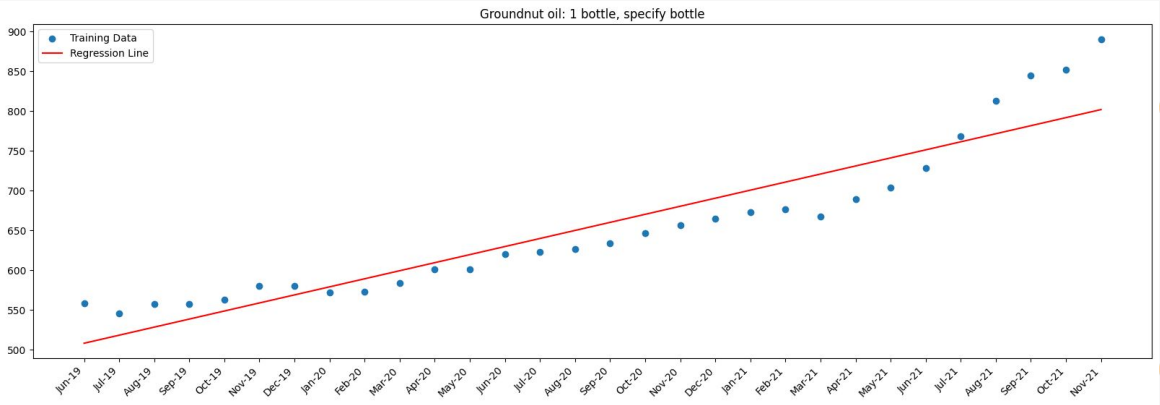
Which linear model is better

?

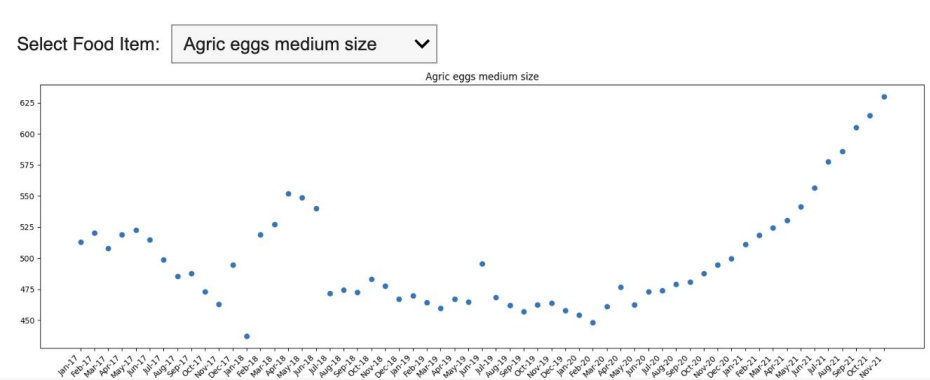
(A)



(B)



Nigerian food item prediction practise Notebook



Practise Notebook: https://colab.research.google.com/drive/1lS_k0gfkzz8lXZqA9WscHuh9QJsEqDcG?usp=sharing#scrollTo=3a4aSRo6T8hi

Linear Regression: loss function

To measure the difference between the predictions $\hat{y}$ and the actual values (labels) y we use the **Mean Squared Error** (MSE)

$$\mathcal{L}(w) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

where

is the number of training examples

is the loss function

Linear Regression: loss function

To measure the difference between the predictions $\hat{y}$ and the actual values (labels) y we use the **Mean Squared Error** (MSE)

$$\mathcal{L}(w) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

where

is the number of training examples

is the loss function

Remember?

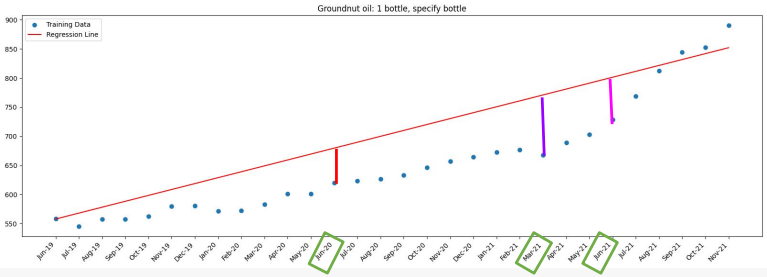
$$c_{i,j} = \sum_{l=1}^{n} a_{il} b_{lj}$$

Linear Regression: loss function

- **Loss** is a penalty for a *bad* prediction - it is a number indicating how **wrong** the model's prediction was on a single data point. If the model's prediction is perfect, the loss is *zero*, otherwise, the loss is *non-zero*

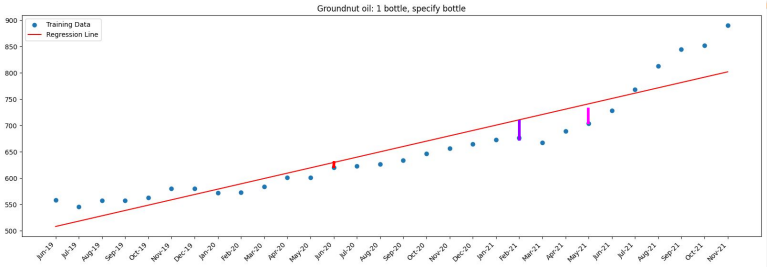
$$\mathcal{L}(w) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

- Training a linear regression model means learning a *set of* values for the weights from labelled examples that reduces the loss



$\mathcal{L}(w)$

Big



$\mathcal{L}(w)$

Small

Updating your model: finding the optimal parameter

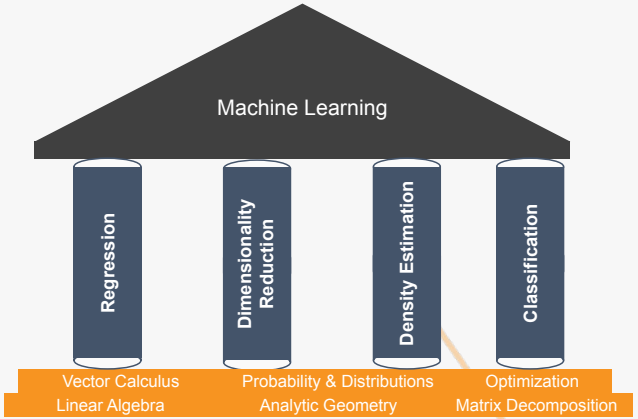
$$\mathcal{L}(w) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

substitute $\hat{y} = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$

$$\mathcal{L}(w) = \frac{1}{m}\sum_{i=1}^{m}(y_i - (w_0 + w_1x_{i1} + w_2x_{i2} + \cdots + w_nx_{in}))^2$$

The goal is to find $w_0, w_1, w_2, \cdots, w_n$ that minimises $\mathcal{L}(w)$

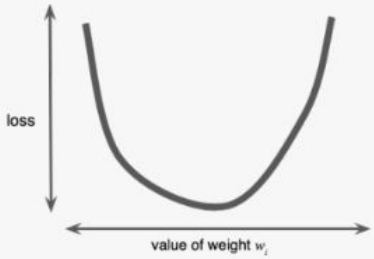
Finding the optimal parameter



We can achieve this in two ways:

- **Closed form solution (normal equation)**
- **Gradient descent optimisation**

$$w = (X^T X)^{-1} X^T y$$



That’s it for today.

What did we *not* cover?

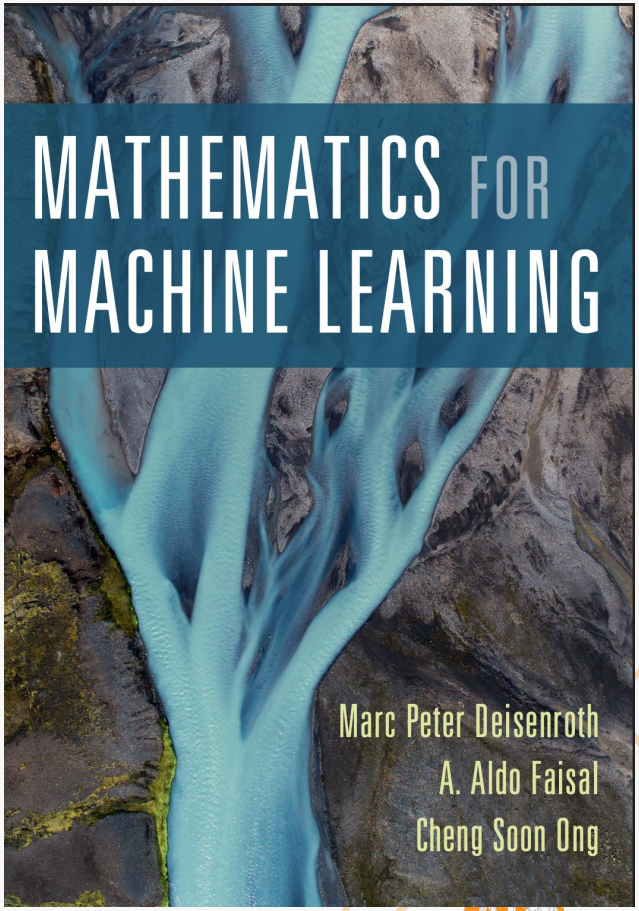
- Alot!

Checkout the textbook on Mathematics for Machine Learning.

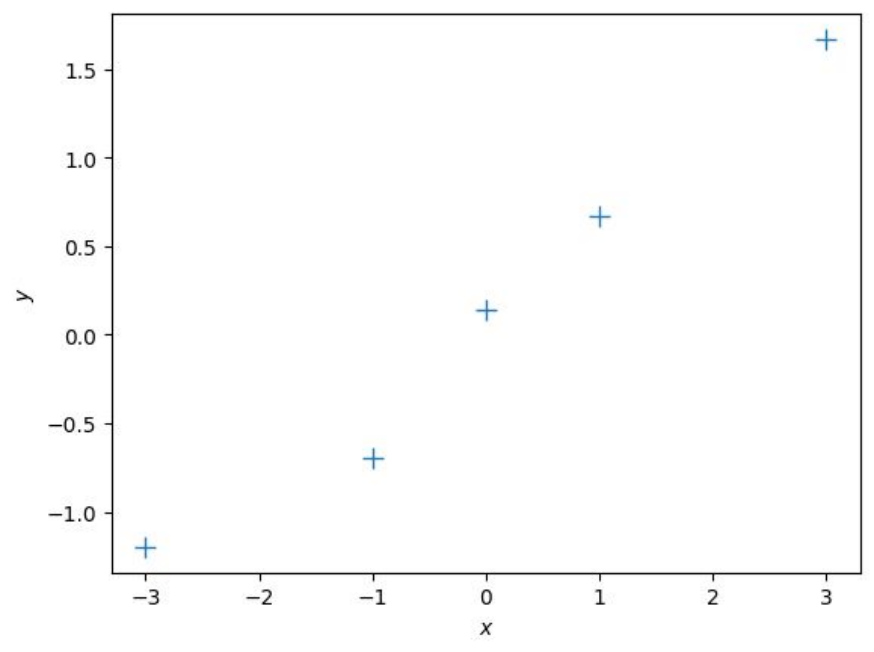
For a tailored Machine Learning Curriculum

Checkout:

https://github.com/AISaturdaysLagos/cohort_structure



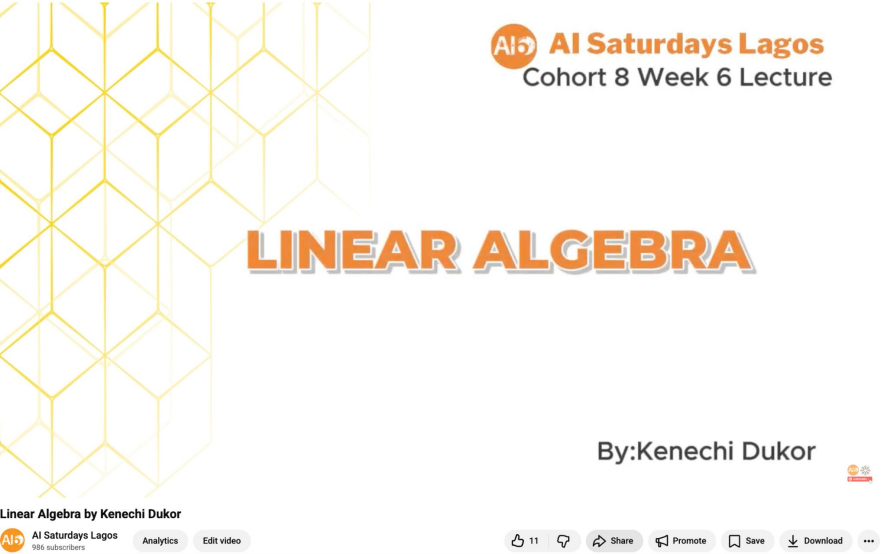
Linear Algebra and Regression Practise Notebook from the book

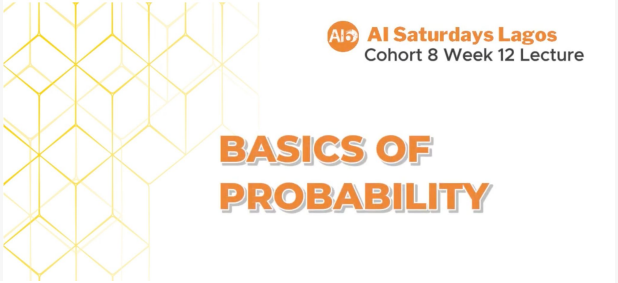


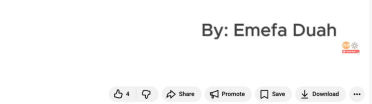
Practise Notebook:

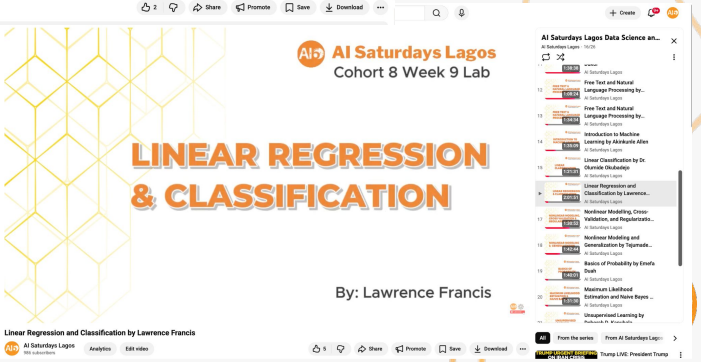
https://github.com/mml-book/mml-book.github.io/blob/master/tutorials/tutorial_linear_regression.ipynb

Our machine learning lectures









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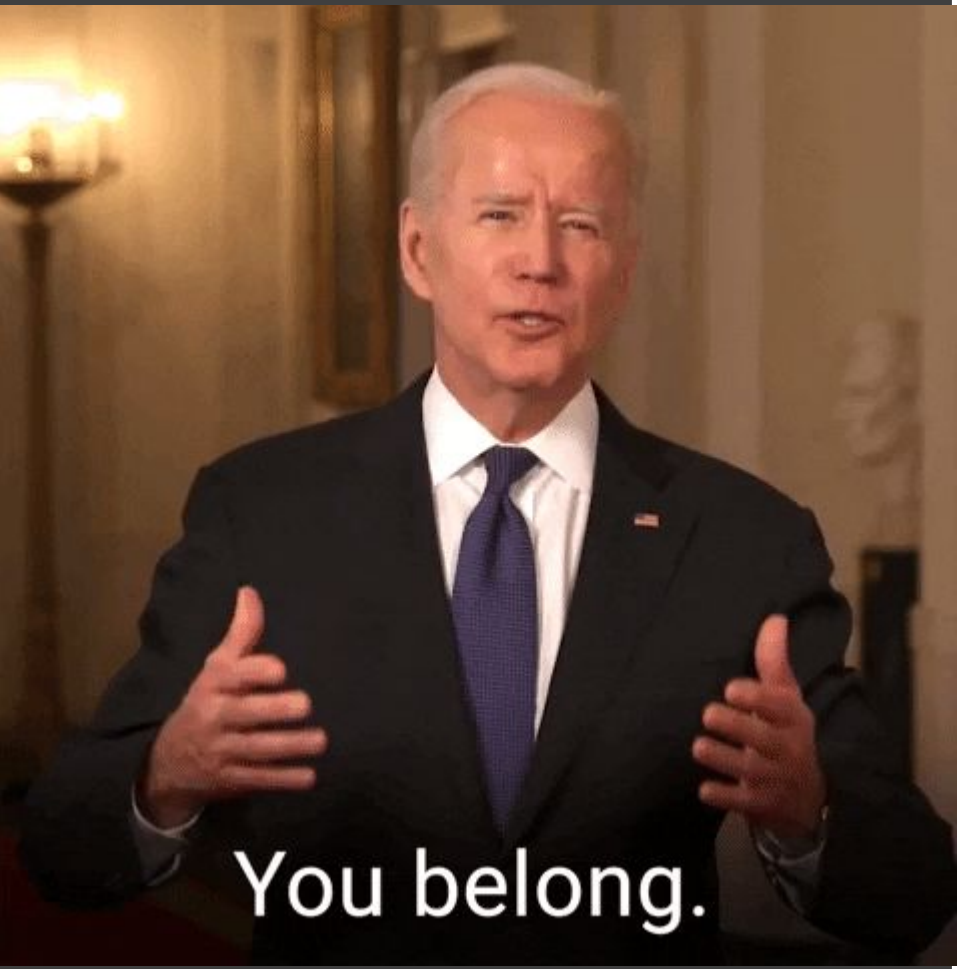
Feedback Form

https://forms.gle/Tu66yWdco2exwC4J9



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